

Layer Collection

The `ABS_LayerCollection` is a Scriptable object holding the necessary layers for the Advanced Building System's scripts.



- Layer Of Building Element
 - Only 1 layer can be set to ensure that the system works properly!
 - This layer will be used to identify the `ABS_BuildingElements`. Make sure your building elements are on this layer.
- Layer Of Player
 - Only 1 layer can be set to ensure that the system works properly!
 - It is used to identify the Player.
 - It is used for blocking the elements during the build process if they are colliding with the player.
 - Also it is used for the PreBuilt Building Element feature.
- Layer Of Ground
 - It is used to identify the ground.
 - For example during the align to ground kind of features and underground check validation.
- Blocking Layer
 - Work similar to the Layer Of the Player but it is only used for blocking the elements if they are colliding with `GameObjects` from those layers during the build process.
- Raycast hit Layers
 - It is used for the raycast. What can be hit by the raycast.
 - Usually it is the mix of the Ground and the `BuildingElement` Layers.

BuildingElementLink

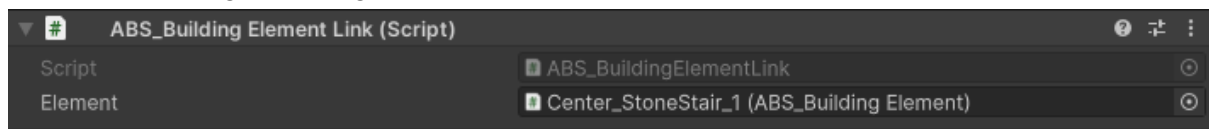
The ABS_BuildingElementLink is a script that was created to fix a raycast bug of the destroy system. The destroy functionality uses raycast to find the elements that the player is looking at. If the raycast hits something the algorithm checks if it has an ABS_BuildingElement component.

Problematic cases:

- If the element has a child object that has a collider that is bigger than the collider on the GameObject with the ABS_BuildingElement (so the parent object) then the algorithm will hit your child's collider and can not find the building element component.
- Or if the child has an exactly the same collider than the building element's collider then something like a "Z fight" effect will happen and the raycast will sometimes hit the parent and sometimes the child.

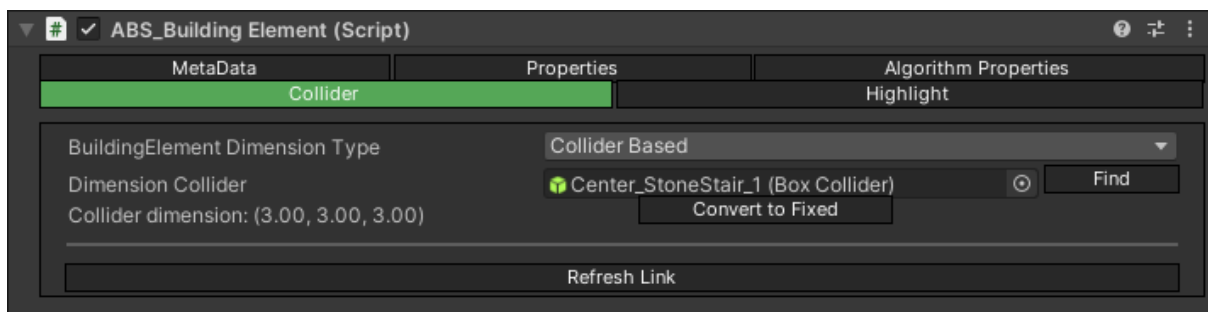
Both cases are problematic because the player is looking at a building element but the raycast still can not find the proper object with the component.

For fixing the bug this script has been created. This script is basically just holding a reference to the original ABS_BuildingElement component. The algorithm next to the ABS_BuildingElement will check if it has an ABS_BuildingElementLink component and if it has a link, it can get the original element's script.



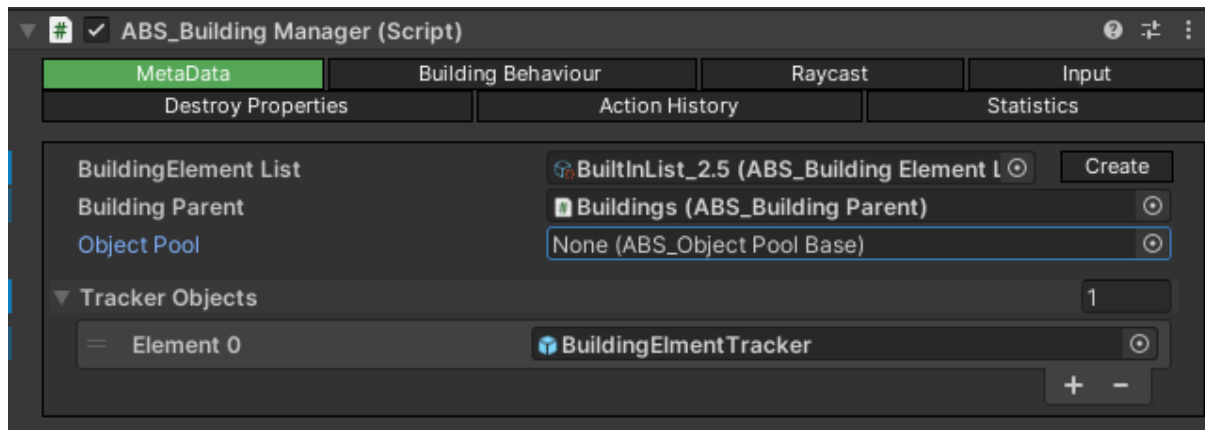
The recommendation is that every child object of a ABS_BuildingElement that has a collider that can be hit by the raycast (based on the raycast settings) has to have a ABS_BuildingElementLink component with a reference to the original building element.

This link is important to ensure that the destroy feature can work properly. And because of that the ABS_BuildingElementCreator tool automatically adds these links to the created building elements. Also the ABS_BuildingElement's UI has an option for refreshing the links. Under the Collider tab with the Refresh Link button the links can be refreshed automatically so the developers should not add them by hand.



ObjectPool/Buffer

The Advanced BuildingSystem can support custom object pools that can be provided to the BuildingManager on its UI. The developer can provide an object pool inherited from the ABS_ObjectPoolBase class. This is not a mandatory property so if it is null then the ABS_BuildingManager will instantiate and destroy the building element but if it is provided the manager will use it to increase the performance.



The ABS_ObjectPoolBase is an abstract class of the following functions that should be implemented.

```
1 //*****
2 // Dependencies: System
3
4 // Dependencies: Unity
5 using UnityEngine;
6
7 // Dependencies: REST
8
9 //*****
10
11 namespace REST.AdvancedBuildSystem
12 {
13     public abstract class ABS_ObjectPoolBase : MonoBehaviour
14     {
15         public abstract ABS_BuildingElement Get(ABS_BuildingElement p_Element);
16         public abstract void GiveBack(ABS_BuildingElement p_Element);
17         public abstract void Release(ABS_BuildingElement p_Element);
18     }
19 }
```

Get

When the Get function is called by the manager it will provide an element prefab as a parameter. The function should return a reference for an instance of the provided prefab. The return value can not be null!

The ABS_BuildingManager will set the necessary properties of the element like parent, rotation, position, layer etc.

GiveBack

When the GiveBack function is called the manager will provide a reference for an instance of a ABS_BuildingElement that can be put back to the pool.

The parent and the set properties are still in that state as the Get function set earlier. So at least the parent object should be reset by the GiveBack function and disable the object for preventing any building logic issue.

Release

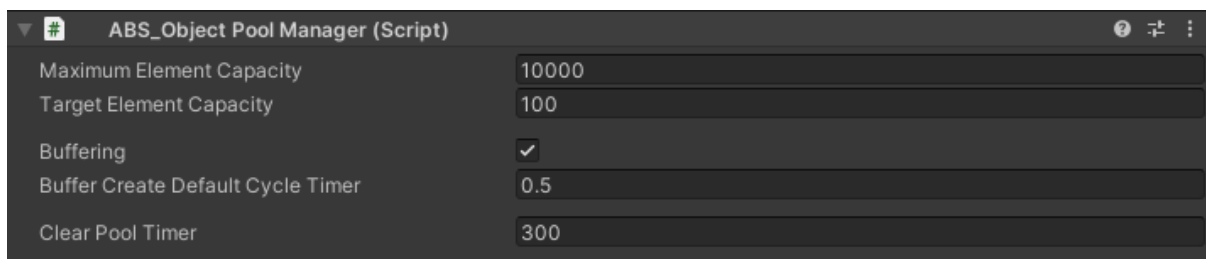
When the Release function is called the manager will provide a reference for an instance of a ABS_BuildingElement that has been placed to the world.

The pool should release its reference to that object because it will never be given back with the GiveBack function so it will not flood the pool's reference counter.

ABS_ObjectPoolManager

The Advanced Building System has its own implementation for the object pool that can be used as well. The ABS_ObjectPoolManager is inherited from the ABS_ObjectPoolBase and implements its required functions so no more action is required than just create an instance of the ABS_ObjectPoolManager and provide it to the BuildingManager on its UI.

This object pool included a buffering logic and a clear logic as well for performance and memory saving purposes.



First of all the manager itself is not the actual object pool, it just creates a pool for every different BuildingElement and forwards the function calls of the ABS_ObjectPoolBase to the right pool based on the BuildingElements. You can see the UI of the pools in action on the image later.

The buffering feature is about to create BuildingElements before it is ever used for performance purposes. A regular object pool when a huge amount of elements is requested and it hasn't any left in the pool then it will create the leftover. It can cause a huge amount of element instantiation in only one frame which can cause fps drop. To prevent this scenario the ABS_ObjectPoolManager's pools automatically create the BuildingElements in a cycle until it reaches the set count. The constant but slow GameObject creation can help to prevent fps drops. It works in such a way in the Update function of the pool the deltatime is constantly checked and when it's reached the timer it will create an element and put it into the pool.

Properties

Maximum Element Capacity

The maximum element counts what can be held by the pools separately. If the limit is reached the pool will return null for the Get function calls. The BuildingManager when it gets a null element it will instantiate the element itself in a regular way.

Target Element Capacity

The target element count of the buffering feature. The buffer cycle will create elements only when the inactive element count (elements in the pool) is smaller than this value. When it has been reached the buffer will stop creating the elements.

Buffering

Boolean value which means that the buffering feature is enabled or not.

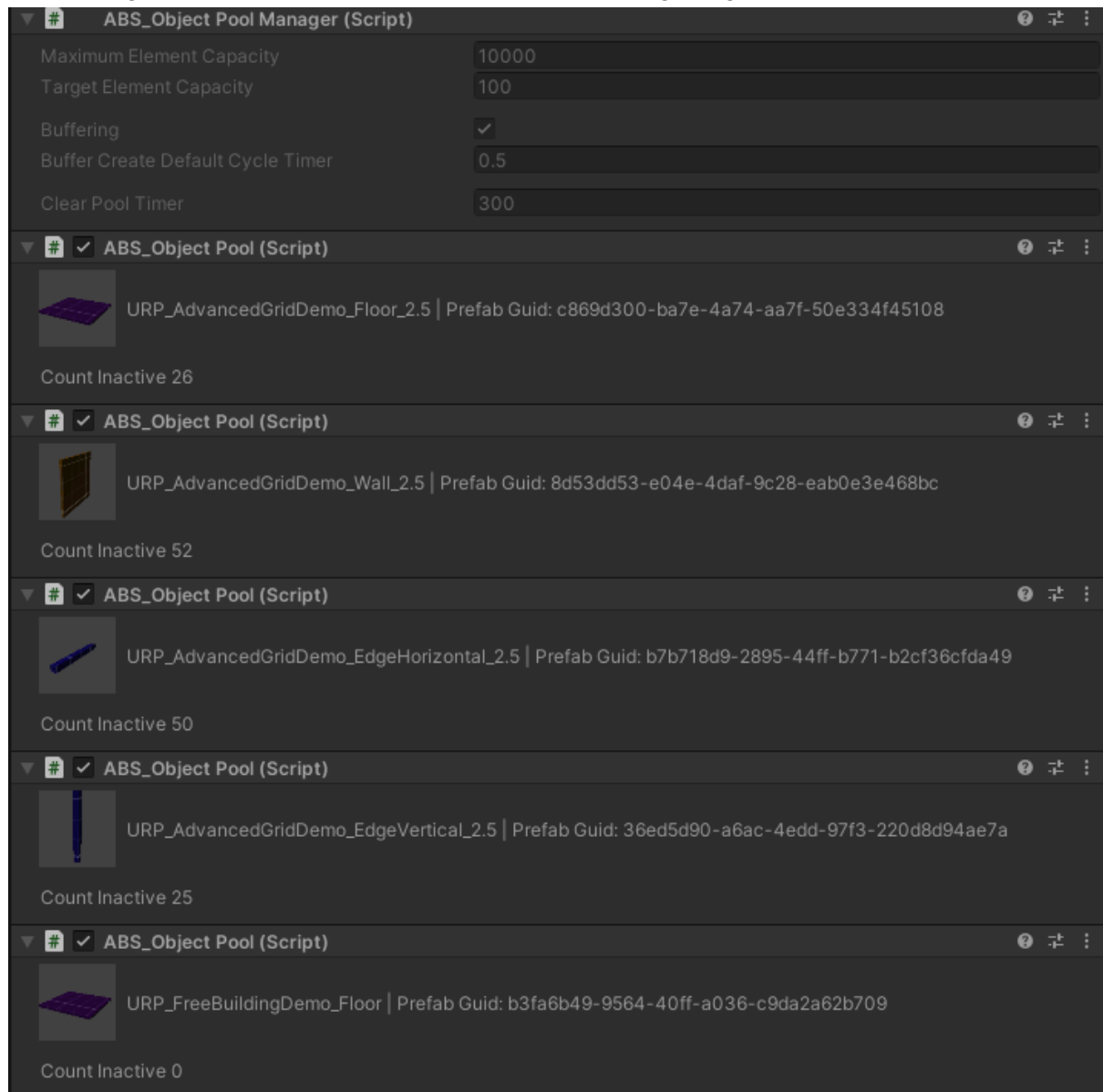
Buffer Create Default Cycle Timer

How much time should be waited between the element creation during the buffering cycle.

Clear Pool Timer

The pools will be cleared (destroyed with its inactive elements) if they didn't get any function calls until the set timer. The Get, GiveBack, Release function calls reset the timer.

The Manager and it's pools can be seen on the following image in action:

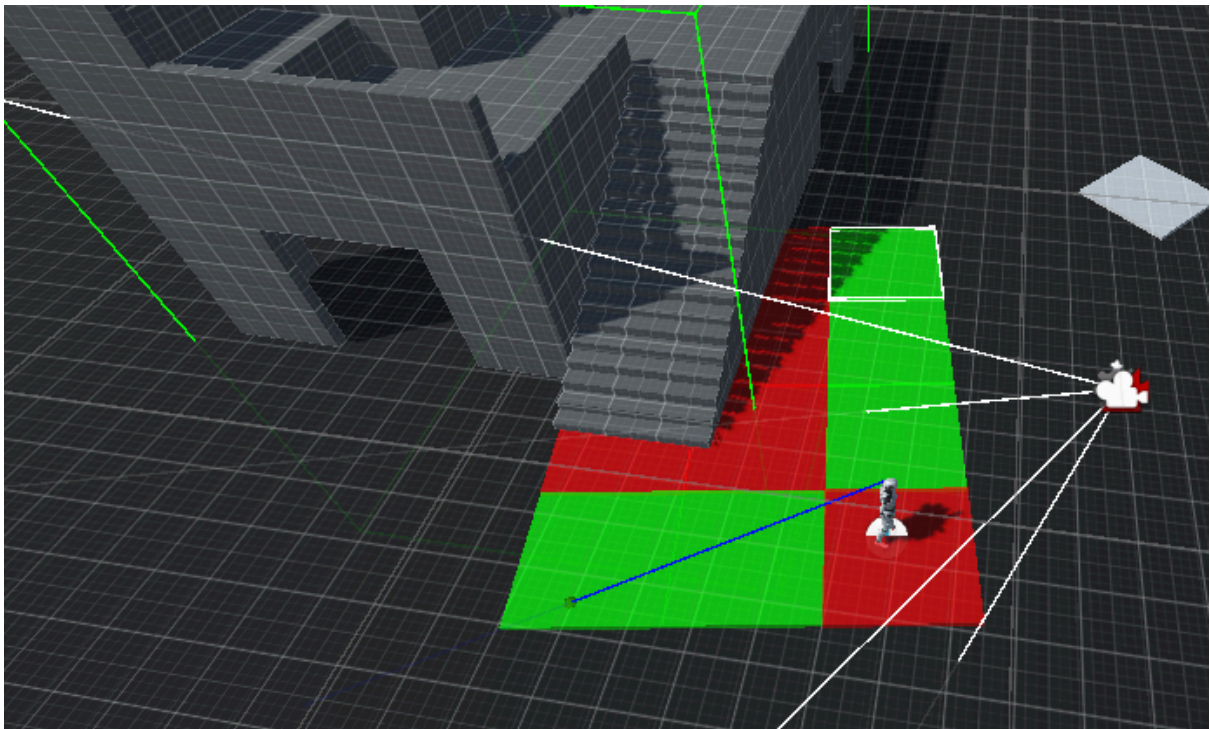


Building Area

The goal of the Building Area Feature is providing the possibility to the developers to create areas in the scene where they can manage the building of the elements. So basically it gives more possibility to the developers to create specific scenarios and manage the behavior of the Advanced Building System like:

- An element can be built near to something, for example an electrical element can be only built near to an electric source.
- Some elements can not be built near to something, for example a wooden element can not be built near the fire.

The main functionality of the BuildingAreas to allow or deny the building of the element.

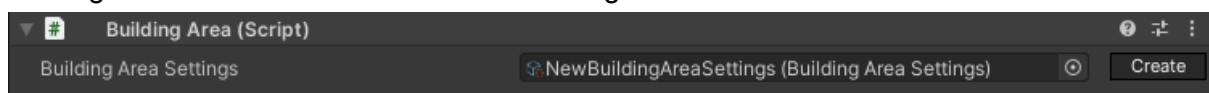


The Building Area created by the following components:

- Building Area (Script component)
- Building Area Settings (Scriptable Object)
- Building Area Ruleset (Scriptable Object)

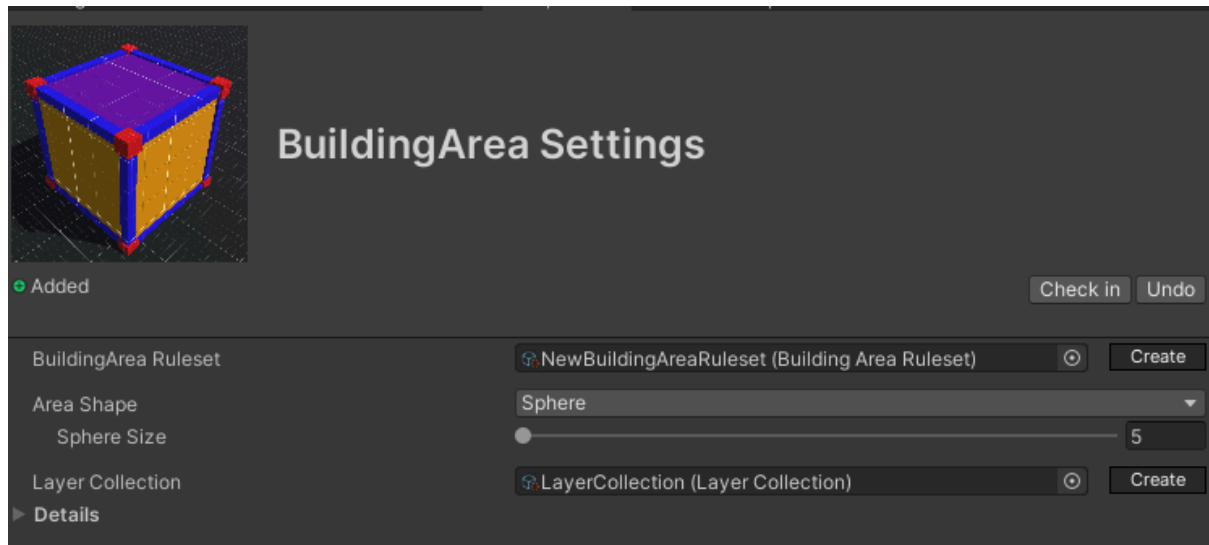
Building Area Component

The script component that is held by the GameObject's should have special area features. In the Awake function the Area will create a trigger collider and allow or deny the element's building inside that collider based on the settings.



Building Area Settings

The goal of the settings scriptable object is to give the possibility to reuse the same area in different situations so the developer doesn't have to set the same properties everytime.



BuildingArea Ruleset : The rules of the area. The allow and deny functionality based on that ruleset.

Area Shape : The shape of the area's collider. Currently 2 shapes are supported.

- Box
- Sphere

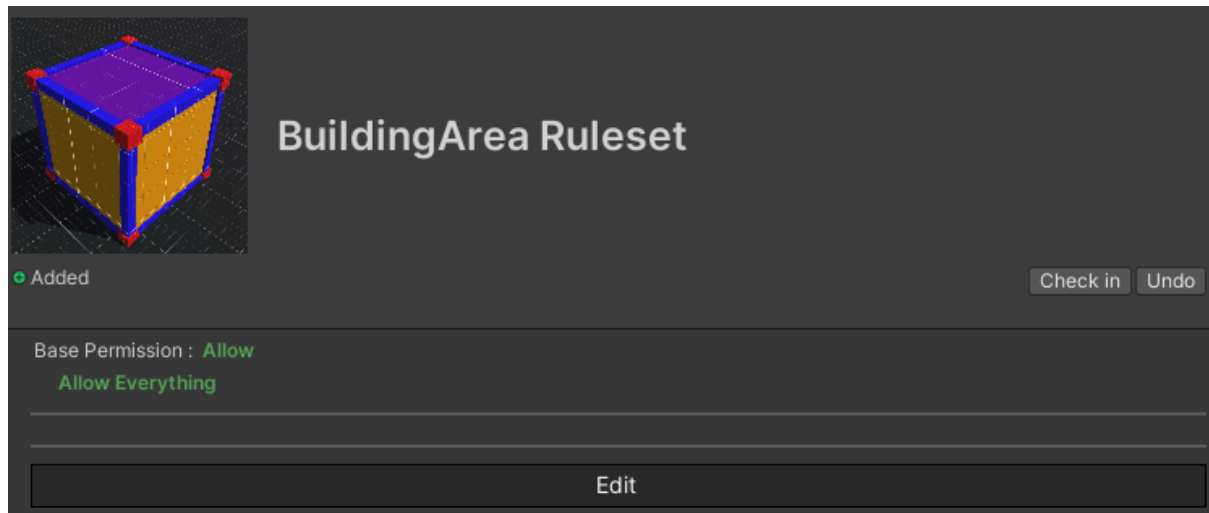
Sphere Size (If the shape is sphere) : The size/radius of the sphere collider created by the Building Area.

Box Size (If the shape is box) : The size of the box collider created by the Building Area in Vector3.

Layer Collection : The layer collection used by the Area.

Building Area Ruleset

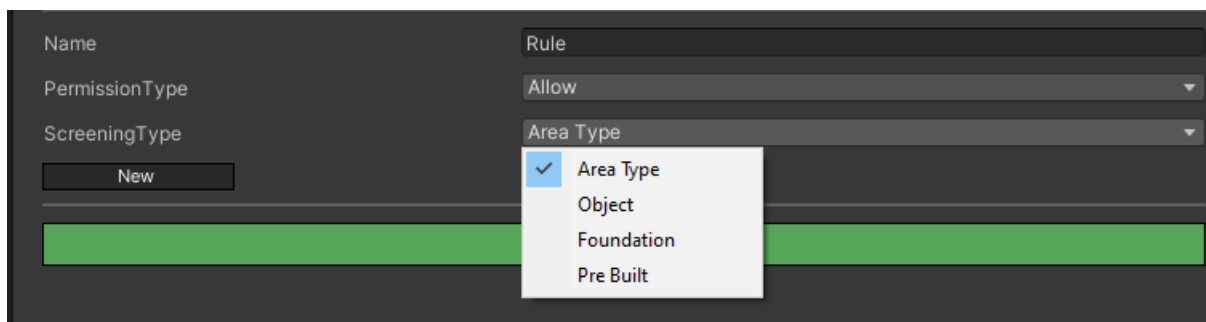
The goal of the ruleset scriptable object is to give the possibility to reuse the ruleset and give the possibility to use the same ruleset with different shaped areas on different areas. The goal is to give the maximum reusability and customizability to the Area feature of the Advanced Building System.



Base Permission: The base permission which will apply on every element what don't included by any of the rules. For example if the Base Permission is Deny and none of the rules said anything about the element (not allowed or denied that element) the building of that element will be blocked inside the area.

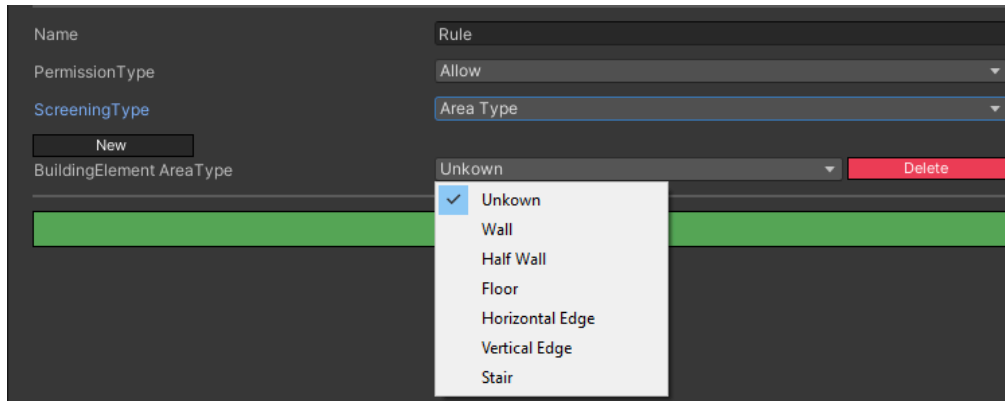
BuildingArea Rules have a PermissionType and a ScreeningType.

- Name
 - The main purposes are to identify the rule itself and make it easier to work with by the developer.
- Permission Type
 - Allow or Deny the elements that fit into the requirements of the rule.
- Screening Type
 - What should the Area check on the element to decide that it should allow or deny based on the Permission Type.



Area Type

The BuildingElement Area Type is an enum value that is defined in the ABS_BuildingElementTypes. You the developer can change these enum values to create your logic. This is meant to identify the elements based on a fully independent type.



Note that every time when the AdvancedBuildingSystem asset is reimported this document can be overridden so ensure that you have a backup from your changes or just do not import this file again.

```
1 //*****
2 // Dependencies: System
3
4 // Dependencies: Unity
5
6 // Dependencies: REST
7
8 //*****
9
10 namespace REST.AdvancedBuildSystem
11 {
12     public enum ABS_BuildingElementAreaType
13     {
14         Unkown,
15         Wall,
16         HalfWall,
17         Floor,
18         HorizontalEdge,
19         VerticalEdge,
20         Stair
21     }
22
23     public enum ABS_BuildingElementSnapPointType
24     {
25         Unkown,
26         Wall,
27         HalfWall,
28         Floor,
29         TriangleFloor,
30         HorizontalEdge,
31         VerticalEdge,
32         Corner,
33         Stair
34     }
35 }
```

ABS_Building Element (Script)

MetaData Properties Algorithm Properties

Collider Highlight

Basics

Pre Built ☐

Foundation ☐

Indestructible ☐

Area Type Floor

Behaviour

Snap To PreBuilt Final Element ☐

Should Snap To Foundation ☐

Should Allowed By Area ☒

Should Override ☐

DragBuilding

Drag Building Enabled ☒

Drag Building Behaviour Filled Up

Enabled Drag Building On X Axis ☒

Enabled Drag Building On Z Axis ☒

Object

If the Rule is set as an Object a set of BuildingElements can be defined. The defined elements will be denied or allowed by the Area.

Name Rule

PermissionType Allow

ScreeningType Object

New

BuildingElement Object	URP_AdvancedGridDemo_Corner_2.5 (Building E)	Delete
BuildingElement Object	URP_AdvancedGridDemo_Floor_2.5 (Building Ele)	Delete

Save

Foundation

It simply denies or allows the building of the elements that are signed as Foundation.

Name	Rule
PermissionType	Allow
ScreeningType	Foundation
Save	

PreBuilt

Just like the Foundation logic denies or allows the building of the elements that are signed as PreBuilt.

Name	Rule
PermissionType	Allow
ScreeningType	Pre Built
Save	

The Relationship of the Rules.

In the Editor view you can move the rules Up and Down or Top and Bottom.

It is important because the order of the checks are from top to bottom.

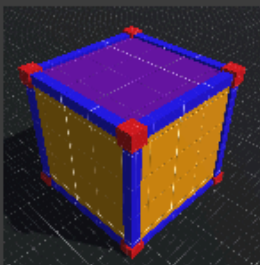
The area's behavior is that:

- If the rule is related to the element and allows it the element will be allowed.
- If the rule is related to the element and Denies it the element will be denied.
- If the rule is not related to the element then the next rule will be checked.

So if the rule is related then the element will be allowed or denied immediately when the rule says any of the two options. For example if the first rule allows the element and the second one denies it the element will be allowed.

Again, the order of the check is from top to bottom.

If all of the rules are unrelated to the element then the base permission will be applied.



BuildingArea Ruleset

Checkout

Base Permission : **Deny** Allow
Deny Everything

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Allow**

CollectionType : AreaType

(1) Wall

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Deny**

CollectionType : Object

0 :  URP_AdvancedGridDemo_Corner_2.5 | cc116ff3-a3a1-47d8-9014-4abbf0beed56

1 :  URP_AdvancedGridDemo_Floor_2.5 | 91ac1293-08c9-4c58-b080-19a8f9108bce

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Allow**

CollectionType : Foundation

▼ Rule

Top Up Down Bottom

Edit Duplicate Delete

PermissionType : **Allow**

CollectionType : PreBuilt

New Entity

Save